(a) Lapse minimum

$\min_n N_n$

1.0
0.9
0.8
0.7
0.6
0.5

Full (staggered)
Proxy (smooth)

0.0 0.5 1.0 1.5 2.0

$\beta_0 t_0$

(b) Bipartition MI

$\alpha_S = 1/4$

α_S

0.25
0.20
0.15
0.10
0.05
0.00

0.0 0.5 1.0 1.5 2.0

$\beta_0 t_0$

(c) Schwarzschild radius

Proxy (smooth)
Full (staggered)

r_s/a

2.0
1.5
1.0
0.5
0.0
−0.5

0.0 0.5 1.0 1.5 2.0

$\beta_0 t_0$

(d) Lattice staggering (full equation)

Odd-even stagger amplitude

10^0
10^{-1}
10^{-2}
10^{-3}

0.0 0.5 1.0 1.5 2.0

$\beta_0 t_0$